Supplementary Information

Online Resource 2: item-level statistics, full model output and an exploratory analysis

Article: *Critical engagement and generative AI dependency: a test of the protective-factor assumption*
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This resource reports (a) item-level descriptive statistics and response distributions (Table S1); (b) the correlations among all items and covariates used in the analyses (Table S2); (c) the full estimates of the measurement model (Table S3); (d) every regression estimate (Table S4) and the fit indices (Table S5) of the primary model and each robustness model; (e) the complete common-method diagnostics (Tables S6 and S7); and (f) an exploratory analysis of the indirect association between critical engagement and AI dependency through perceived benefits (Table S8).

Item wording is given in Appendix Table A1 of the article. The robustness specifications R1–R10 are described in Section 3.4 and Table 5 of the article. All values were produced by the R script provided as Online Resource 1 (Sections 9 and 12), and each is rounded once from the unrounded estimate. $N = 552$ unless stated otherwise.

Table S1 Item-level descriptive statistics and response distributions

						Responses at each scale point (%)						
Item	*n*	*M*	*SD*	Skew.	Kurt.	1	2	3	4	5	6	7
Critical engagement												
Q8 Starting point for own ideas	552	4.82	1.73	−0.39	−0.78	4.0	7.4	10.7	20.7	17.8	16.3	23.2
Q9 Cross-checking with other sources	552	4.93	1.66	−0.46	−0.70	2.9	6.3	10.7	20.8	14.9	23.4	21.0
AI dependency												
Q6 Reliance for decisions	552	4.18	1.55	−0.23	−0.32	7.4	7.1	12.9	32.1	21.4	12.0	7.2
Q7 AI instead of own problem-solving	552	4.26	1.55	−0.13	−0.48	5.3	8.0	15.9	27.9	21.4	12.9	8.7
Q10 Acceptance without verification	552	4.34	1.78	−0.31	−0.77	9.2	9.2	9.4	23.0	21.2	15.0	12.9
Control variables												
Q5 Frequency of use (1–5)	552	4.19	0.86	−0.93	0.61	0.9	2.4	17.0	36.6	43.1	–	–
Q4 AI-use experience (1–4)	552	3.34	0.81	−1.19	0.93	4.3	8.3	36.1	51.3	–	–	–
Perceived benefits (covariate in R2)												
Q17 Saves time	552	5.55	1.51	−1.10	0.63	2.5	1.8	7.6	9.6	16.1	28.8	33.5
Q18 Confidence in problem-solving	552	4.71	1.72	−0.56	−0.39	7.2	5.8	7.4	19.4	25.7	17.2	17.2
Q19 Learning efficiency	552	5.05	1.50	−0.58	0.03	2.9	3.4	6.5	19.6	29.3	16.8	21.4
Q20 Satisfaction with learning	552	4.60	1.42	−0.44	0.04	3.6	4.2	11.4	24.5	31.0	16.8	8.5
Q21 Collaborative learning tool	552	5.20	1.47	−0.79	0.35	3.1	1.6	7.1	17.0	24.1	26.3	20.8
AI-induced remorse (marker variable)												
Q11 Fear of weakened thinking	552	5.30	1.74	−0.85	−0.17	4.7	3.8	7.1	14.1	17.2	17.6	35.5
Q12 Reduced recall	552	4.70	1.78	−0.51	−0.57	8.0	5.1	8.9	21.2	19.6	18.1	19.2
Q14 Remorse for avoiding challenges	552	4.29	1.69	−0.21	−0.66	7.6	8.7	12.3	26.4	19.7	13.8	11.4
Q15 Regret over dependency	552	4.16	1.76	−0.15	−0.77	10.0	9.8	12.1	25.0	20.5	11.1	11.6
Q16 Regret after tasks	552	4.17	1.70	−0.24	−0.64	9.6	10.0	9.2	26.8	23.7	10.7	10.0
Academic performance (covariate in R8)												
CGPA (0–4)	550	3.34	0.35	−0.33	−0.41	–	–	–	–	–	–	–

Note. Item wording is given in Appendix Table A1 of the article. Seven-point items range from 1 (lowest) to 7 (highest agreement or reliance); Q5: 1 = *never* to 5 = *always*; Q4: 1 = less than 6 months to 4 = more than 2 years; CGPA: self-reported, 0–4. Skewness and excess kurtosis as computed by the `describe` function of psych. Percentages may not sum to 100 because of rounding.

Table S2 Correlations among all items and covariates used in the analyses

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Q8 Starting point for own ideas																	
2. Q9 Cross-checking with other sources	.46***																
3. Q6 Reliance for decisions	.29***	.16***															
4. Q7 AI instead of own problem-solving	.37***	.25***	.53***														
5. Q10 Acceptance without verification	.27***	.08	.35***	.29***													
6. Q5 Frequency of use (1–5)	.24***	.13**	.29***	.27***	.12**												
7. Q4 AI-use experience (1–4)	.11*	.06	.05	.10*	.00	.25***											
8. Q17 Saves time	.44***	.32***	.27***	.36***	.28***	.13**	.02										
9. Q18 Confidence in problem-solving	.42***	.24***	.26***	.29***	.23***	.26***	.02	.44***									
10. Q19 Learning efficiency	.39***	.35***	.21***	.36***	.28***	.19***	.06	.57***	.61***								
11. Q20 Satisfaction with learning	.36***	.27***	.25***	.37***	.29***	.20***	.08	.43***	.47***	.56***							
12. Q21 Collaborative learning tool	.39***	.34***	.21***	.36***	.25***	.19***	.10*	.63***	.43***	.58***	.60***						
13. Q11 Fear of weakened thinking	.25***	.25***	.03	.18***	.15***	.04	−.04	.37***	.20***	.25***	.14**	.23***					
14. Q12 Reduced recall	.12**	.16***	.07	.09*	.18***	−.01	.05	.23***	.04	.08*	.07	.11*	.59***				
15. Q14 Remorse for avoiding challenges	.07	.04	.01	.08	.09*	−.18***	−.08	.16***	.10*	.07	.09*	.05	.34***	.47***			
16. Q15 Regret over dependency	.10*	.10*	.01	.10*	.15***	−.07	−.07	.17***	.08	.16***	.11*	.08	.32***	.47***	.69***		
17. Q16 Regret after tasks	.06	.07	−.06	.12**	.08	−.19***	−.03	.06	.07	.11**	.05	−.00	.34***	.46***	.61***	.69***	
18. CGPA (0–4)	.15***	.18***	.09*	.13**	−.01	−.01	.08	.10*	.05	.11*	.01	.04	.08*	−.03	.01	−.01	.02

Note. Pearson correlations, $N = 552$ (pairs involving CGPA: $n = 550$). Means, standard deviations and response distributions are given in Table S1. * $p < .05$; ** $p < .01$; *** $p < .001$ (two-tailed, unadjusted).

Table S3 Measurement model: full parameter estimates

Item	*B*	*SE*	*z*	*p*	λ [95% CI]	θ	R^2
Critical engagement (two-factor CFA, Table 2 of the article)							
Q8 Starting point for own ideas	1.00[a]	–	–	–	.88 [.74, 1.03]	0.65	.78
Q9 Cross-checking with other sources	0.56	0.11	5.28	< .001	.52 [.39, .65]	1.99	.27
AI dependency (two-factor CFA)							
Q6 Reliance for decisions	1.00[a]	–	–	–	.71 [.62, .79]	1.20	.50
Q7 AI instead of own problem-solving	1.05	0.12	8.93	< .001	.74 [.64, .84]	1.08	.55
Q10 Acceptance without verification	0.73	0.10	7.28	< .001	.45 [.34, .56]	2.51	.20
Perceived benefits (measurement part of robustness model R2)							
Q17 Saves time	1.00[a]	–	–	–	.73 [.66, .79]	1.08	.53
Q18 Confidence in problem-solving	1.04	0.09	11.81	< .001	.67 [.59, .74]	1.64	.44
Q19 Learning efficiency	1.08	0.07	14.87	< .001	.79 [.74, .84]	0.83	.63
Q20 Satisfaction with learning	0.90	0.07	12.87	< .001	.70 [.64, .77]	1.01	.50
Q21 Collaborative learning tool	1.02	0.06	17.45	< .001	.77 [.71, .83]	0.88	.59

Note. MLR estimates, $N = 552$. B = unstandardised loading with robust standard error; λ = standardised loading with delta-method 95% CI; θ = unstandardised residual variance; R^2 = proportion of item variance explained by the factor. Fit indices are given in Table S5.

[a] Fixed to 1 to set the scale of the factor.

Table S4 Regression estimates for the primary model and every robustness model

Predictor	*B*	*SE*	*z*	*p*	β [95% CI]
Primary model (H1)					
Critical engagement	0.33	0.07	4.90	< .001	.46 [.32, .60]
Frequency of use	0.34	0.08	4.51	< .001	.26 [.16, .37]
AI-use experience	−0.04	0.06	−0.62	.533	−.03 [−.12, .06]
R1 No control variables					
Critical engagement	0.38	0.07	5.21	< .001	.53 [.40, .67]
R2 Adds perceived benefits (latent covariate)					
Critical engagement	0.21	0.08	2.61	.009	.28 [.08, .48]
Perceived benefits	0.32	0.09	3.69	< .001	.34 [.15, .52]
Frequency of use	0.26	0.07	3.87	< .001	.21 [.11, .31]
AI-use experience	−0.03	0.06	−0.44	.663	−.02 [−.11, .07]
R3 Four-item dependency scale including the frequency item					
Critical engagement	0.38	0.07	5.40	< .001	.53 [.40, .66]
AI-use experience	0.09	0.07	1.37	.171	.07 [−.03, .17]
R4 Reliability-corrected single-indicator critical engagement					
Critical engagement (single indicator)	0.43	0.06	7.30	< .001	.47 [.33, .60]
Frequency of use	0.34	0.08	4.51	< .001	.26 [.16, .37]
AI-use experience	−0.04	0.06	−0.61	.540	−.03 [−.11, .06]
R5 Items Q8 and Q9 entered together					
Q8 Starting point for own ideas	0.24	0.03	7.15	< .001	.37 [.27, .47]
Q9 Cross-checking with other sources	0.05	0.04	1.21	.227	.07 [−.04, .18]
Frequency of use	0.37	0.07	5.10	< .001	.29 [.19, .38]
AI-use experience	−0.03	0.06	−0.51	.608	−.02 [−.11, .06]
R5 Item Q9 entered alone					
Q9 Cross-checking with other sources	0.16	0.04	3.94	< .001	.23 [.10, .35]
Frequency of use	0.48	0.08	5.92	< .001	.35 [.25, .45]
AI-use experience	−0.02	0.06	−0.27	.788	−.01 [−.10, .07]
R6 Excluding straight-lining respondents					
Critical engagement	0.31	0.07	4.37	< .001	.44 [.29, .59]
Frequency of use	0.35	0.08	4.37	< .001	.27 [.16, .38]
AI-use experience	−0.02	0.06	−0.37	.713	−.02 [−.11, .07]
R7 Ordinal estimation (WLSMV)					
Critical engagement	0.35	0.05	7.15	< .001	.47 [.37, .56]
Frequency of use	0.22	0.04	4.97	< .001	.26 [.16, .36]

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Table S4 (continued)

Predictor	*B*	*SE*	*z*	*p*	β [95% CI]
AI-use experience	−0.03	0.04	−0.84	.403	−.04 [−.12, .05]
R8 Adds self-reported CGPA					
Critical engagement	0.36	0.07	5.20	< .001	.47 [.33, .61]
Frequency of use	0.33	0.08	4.28	< .001	.25 [.15, .36]
AI-use experience	−0.04	0.06	−0.70	.487	−.03 [−.12, .06]
CGPA	0.16	0.15	1.02	.309	.05 [−.04, .14]
R9 Dependency measured by Q6 and Q7 only					
Critical engagement	0.33	0.07	4.94	< .001	.47 [.33, .60]
Frequency of use	0.32	0.08	3.92	< .001	.26 [.15, .37]
AI-use experience	−0.02	0.06	−0.32	.751	−.01 [−.11, .08]
R10 Outcome: acceptance without verification (Q10)					
Critical engagement	0.32	0.06	5.28	< .001	.28 [.17, .38]
Frequency of use	0.11	0.09	1.24	.215	.05 [−.03, .14]
AI-use experience	−0.10	0.08	−1.16	.246	−.04 [−.12, .03]

Note. Outcome: AI dependency (latent), except in R3 (four-item factor including Q5), R9 (two-item factor of Q6 and Q7) and R10 (Q10, observed). MLR estimates with robust standard errors, except R7 (WLSMV, items treated as ordinal). Standardised 95% CIs by the delta method. $n = 552$, except R6 ($n = 533$) and R8 ($n = 550$). The specifications are described in Section 3.4 and Table 5 of the article; their fit indices are in Table S5.

Table S5 Fit indices and explained variance for the primary model and every robustness model

Model	*n*	χ^2 (*df*)	*p*	CFI	TLI	RMSEA [90% CI]	SRMR	R^2
Two-factor CFA (Table 2)	552	15.60 (4)	.004	.972	.931	.079 [.040, .121]	.031	–
Primary model (H1)	552	24.34 (10)	.007	.976	.949	.051 [.026, .078]	.027	.34
R1 No control variables	552	15.60 (4)	.004	.972	.931	.079 [.040, .121]	.031	.29
R2 Adds perceived benefits (latent covariate)	552	166.95 (46)	< .001	.929	.898	.075 [.063, .088]	.043	.44
R3 Four-item dependency scale including the frequency item	552	53.05 (12)	< .001	.929	.876	.080 [.059, .102]	.048	.30
R4 Reliability-corrected single-indicator critical engagement	552	15.25 (6)	.018	.979	.948	.052 [.020, .086]	.024	.34
R5 Items Q8 and Q9 entered together	552	23.66 (8)	.003	.959	.924	.061 [.033, .090]	.027	.30
R5 Item Q9 entered alone	552	13.45 (6)	.036	.979	.958	.047 [.011, .081]	.025	.19
R6 Excluding straight-lining respondents	533	24.64 (10)	.006	.973	.942	.053 [.027, .079]	.028	.32
R7 Ordinal estimation (WLSMV)	552	37.91 (10)	< .001	.975	.948	.071 [.048, .096]	.030	.34
R8 Adds self-reported CGPA	550	32.34 (13)	.002	.968	.931	.053 [.030, .076]	.030	.36
R9 Dependency measured by Q6 and Q7 only	552	8.04 (5)	.154	.994	.983	.032 [.000, .071]	.016	.34
R10 Outcome: acceptance without verification (Q10)	552	4.69 (5)	.455	1.000	1.003	.000 [.000, .060]	.020	.09

Note. For MLR models, χ^2 is the Yuan–Bentler scaled statistic and CFI, TLI and RMSEA are the robust versions; for R7 (WLSMV), the scaled-and-shifted statistic and the scaled indices are reported. R^2 refers to the outcome of each model. The RMSEA is unreliable in models with few degrees of freedom.

Table S6 Common-method variance: fit of the compared models

Model	χ^2 (df)	p	CFI	TLI	RMSEA [90% CI]	SRMR	Latent r [95% CI]
Two-factor CFA, five focal items (Table 2)	15.60 (4)	.004	.972	.931	.079 [.040, .121]	.031	.53 [.40, .67]
Single-factor CFA, five focal items	87.26 (5)	< .001	.815	.630	.182 [.150, .216]	.080	–
Four-factor CFA, 15 seven-point items	295.48 (83)	< .001	.918	.896	.076 [.067, .085]	.069	.58 [.46, .69]
Four-factor CFA with ULMC	276.54 (82)	< .001	.929	.909	.071 [.062, .080]	.059	.01 [−.48, .50]
Two-factor CFA with CFA marker factor	103.72 (32)	< .001	.951	.931	.070 [.055, .085]	.062	.52 [.38, .66]

Note. MLR estimates, $N = 552$; Yuan–Bentler scaled χ^2 and robust fit indices. Latent r is the correlation between critical engagement and AI dependency. The four-factor CFA comprises critical engagement, AI dependency, perceived benefits and AI-induced remorse (residual covariance of Q11 and Q12 freed). The ULMC loads all 15 items on a method factor with equal unstandardised loadings, uncorrelated with the trait factors. The CFA marker model adds the five remorse items as a marker factor, with equal unstandardised loadings on the five focal items and no correlation with the focal factors.

Table S7 Common-method variance: further diagnostics

Diagnostic	Result
Harman: variance explained by the first unrotated component (16 items)	29.9%
Single- versus two-factor model of the five focal items: scaled $\Delta\chi^2(1)$	91.72, $p < .001$
ULMC: standardised method loadings (range)	.45 to .58
ULMC: mean share of item variance due to the method factor	26%
ULMC: change in robust CFI	.011
Lindell–Whitney: observed r(CE, AD), scale means	.36
Lindell–Whitney: marker correlation r_M	.15
Lindell–Whitney: adjusted r(CE, AD)	.25, $t(549) = 5.96$, $p < .001$
CFA marker: standardised method loadings on the focal items (range)	.09 to .11
CFA marker: mean share of focal-item variance due to the marker factor	1%

Note. $N = 552$. Harman's test uses the 15 seven-point items and the frequency item. The marker variable is AI-induced remorse; r_M is the smaller of its correlations with critical engagement (.19) and AI dependency (.15), computed from scale means. The method loadings are equal in unstandardised form; their standardised values differ across items.

Table S8 Exploratory analysis: indirect association between critical engagement and AI dependency through perceived benefits

Path	B	SE	z	p	β [95% CI]	Bootstrap 95% CI for B
A. Perceived benefits as an intervening variable						
Critical engagement → perceived benefits (a)	0.53	0.08	6.75	< .001	.67 [.56, .78]	[0.38, 0.70]
Perceived benefits → AI dependency (b)	0.32	0.09	3.69	< .001	.34 [.15, .52]	[0.13, 0.50]
Critical engagement → AI dependency, direct (c')	0.21	0.08	2.61	.009	.28 [.08, .48]	[0.06, 0.41]
Indirect association ($a \times b$)	0.17	0.05	3.50	< .001	.23 [.10, .36]	[0.08, 0.30]
Total association ($c' + a \times b$)	0.39	0.05	7.06	< .001	.51 [.39, .63]	[0.29, 0.52]
B. Equivalent model with the reverse ordering						
Perceived benefits → critical engagement	0.83	0.07	12.15	< .001	.66 [.55, .77]	–
Critical engagement → AI dependency	0.21	0.08	2.61	.009	.28 [.08, .48]	–
Perceived benefits → AI dependency, direct	0.32	0.09	3.69	< .001	.34 [.15, .52]	–
Indirect association	0.18	0.07	2.58	.010	.19 [.05, .32]	–
Total association	0.50	0.05	10.02	< .001	.52 [.42, .63]	–

Note. Exploratory analysis, not a hypothesis test; $N = 552$. Frequency of use and AI-use experience are controlled in every equation. Both models re-parameterise robustness model R2 and therefore have its fit (Table S5). MLR estimates with robust standard errors; standardised 95% CIs by the delta method. Bootstrap CIs are bias-corrected (5,000 resamples, maximum likelihood; none of the draws was non-admissible) and were computed for Panel A only. In Panel A, the indirect association accounts for 45% of the total association; $R^2 = .48$ for perceived benefits and .44 for AI dependency. Because the two orderings fit the data equally well, the indirect association shows how much of the association is shared with perceived benefits; it does not establish mediation.